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DETAIL SPECIFICATION

PRINTED WIRING ASSEMBLY, CLOCK/DISPLAY DRIVER

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a Clock/Display Driver Printed Wiring Assembly referred to herein as the PWA. This is one of six printed wiring assemblies which comprise the Simplified Test Equipment/Internal Combustion Engine-Reprogrammable (STE/ICE-R) Vehicle Test Meter (VTM) (see Drawing 12259265).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5998

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-810 - Environmental Test Methods and Engineering Guidelines (see 4.2.1).

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

12258780 - Printed Wiring Assembly, Clock/Display Driver.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/IPC J-STD-001 - Requirements for Soldered Electrical and Electronic Assemblies.

(Application for copies should be addressed to the American National Standards Institute (ANSI), 11 West 42nd Street, New York, NY 10036.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.1.

3.2 Design, materials, and manufacturing processes. Design, materials, and manufacturing process selection shall be as specified herein and in applicable referenced specifications, standards, and drawings. Materials shall be uniform and free from imperfections or defects which affect their performance and serviceability. All metallic parts shall be made from corrosion-resistant metals or treated with corrosion-resistant materials. Asbestos and cadmium materials shall not be used in any form in any part of the PWA. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcuries per gram or activity per item equals or exceeds 0.01 microcuries.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Operating requirements. Each PWA shall provide the following functional, operational, and performance capabilities.

3.3.1 Input power. The PWA shall require not more than the current specified in table I.

TABLE I. Input power.

Source	Voltage	Current
P1-5, 6	5 Vdc 1/	285 mA 2/

1/ Vdc = volts direct current.
2/ mA = milliampere.

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3.3.2 Complementary metal oxide semiconductor (CMOS) logic levels. Unless otherwise specified (see 6.2), CMOS logic levels accepted by and output from the PWA shall be:

Input High (Logic Level 1) = 3.5 Vdc minimum (min)
Input Low (Logic Level 0) = 1.5 Vdc maximum (max)
Output High (Logic Level 1) = 4.5 Vdc min
Output Low (Logic Level 0) = 0.5 Vdc max

3.3.3 Input isolation. (NOTE: A "/" following a capitalized signal name denotes logic negation.)

3.3.3.1 Data BUS isolation. Logic levels placed on a BUS input (P1-21 thru P1-28) shall have no effect on nor be affected by logic levels present at any other BUS input.

3.3.3.2 I/O, MRD, and OEN-Q isolation. Logic levels present at any one of P1-29 thru P1-34, P1-36, and P1-38 shall have no effect on nor be affected by logic levels present at any of the other inputs specified in this paragraph.

3.3.4 Display driver outputs. Voltages appearing at a display output (P1-49 through P1-80) shall have no influence on nor be influenced by voltages appearing at any other display output. Display outputs, when pulled up to 5.0 V with a 270 ohm resistor, shall be capable of sinking not less than 14.1 mA when active (logic level 1 at latch output) and not greater than 8.75 mA when inactive (logic level 0 at latch output).

3.3.5 Real time clock.

3.3.5.1 Clock clear. When a logic level 1 is applied to both P1-38 and P1-33, U22 pin 10 shall maintain a logic level 1 while U2 pins 3, 4, 5, 6, 11, 12, 13, 14 and U21 pins 3, 4, 5, 6, 11, 12, 13, and 14 shall maintain a logic level 0 regardless of any other inputs.

3.3.5.2 Q pulse. The Q pulse appearing at U16 pin 13 and P1-42 shall meet the timing restrictions shown in figure 1.

3.3.5.3 Clock count and latch. The real time clock, when cleared and allowed to run for Trun seconds at which time the data shall be latched into U1, U7, U20, and U15 by the Q pulse, shall have the decimal equivalent of the 16-bit binary clock count of (Fin * Trun)/4; -0/+1 where: Fin = the frequency of the clock signal applied to P1-45 and (16384/Fin) < Trun < (262140/Fin). Trun shall be accurate and repeatable to five decimal places.

3.3.6 Automated testing equipment (ATE) test. The PWA shall pass TACOM approved ATE test installed in TACOM approved ATE or equivalent.

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3.3.7 Soldering. Soldering shall meet or exceed the requirements of ANSI/IPC J-STD-001 (see 4.2).

3.4 Interface requirements. Each PWA shall accommodate the following requirements.

3.4.1 PWA overall envelope. Overall envelope of the PWA shall be in accordance with Drawing 12258780.

3.5 Ownership and support requirements.

3.5.1 Interchangeability. Interchangeability tolerances should permit parts, subassemblies and assemblies to be used in their parent assemblies without regard to the source of supply or manufacturer. Parts, subassemblies and assemblies having the full range of dimensions and characteristics permitted by the specification governing the part, subassembly or assembly should be usable as replacement items without selection and without departure from the specified performance guidelines of the parent items.

3.5.2 Identification marking. Unless otherwise specified (see 6.2), identification marking shall be permanent and legible and shall include as minimum the following:

a. Nomenclature “PRINTED WIRING ASSEMBLY, CLOCK/DISPLAY DRIVER”.
b. Military part number.
c. Manufacturer’s CAGE code and name.
d. Manufacturer’s serial numbers.
e. Contract number.
f. Symbol “US”.

3.6 Operating environment requirements. Each PWA shall operate under the following environmental conditions without damage or loss of performance.

3.6.1 Temperature.

3.6.1.1 High temperature (operating). The PWA shall be capable of operation without damage or malfunction at a high temperature of 125 degrees Fahrenheit (°F) (52 degrees Celsius (°C)) (without solar radiation).

3.6.1.2 Low temperature (operating). The PWA shall be capable of operation without damage or malfunction at a low temperature of 20°F (minus (-) 7°C) for ambient temperature below 20°F, the test set shall be operated in a shelter.

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3.6.1.3 High temperature (nonoperating). The PWA shall be capable of being stored and transported in its case without damage at an ambient temperature of 160°F (71°C).

3.6.1.4 Low temperature (nonoperating). The PWA shall be capable of being stored and transported in its case without damage in ambient temperatures of -60°F (-51°C).

3.6.2 Fungus. All nonmetallic materials that are not inherently fungistatic shall function in a fungus environment without adversely affecting the operation/performance of the PWA.

3.6.3 Shock. The PWA shall evidence no degradation of function and meet the requirements of 3.3.1 thru 3.3.6 when subjected to the shocks test.

3.6.4 Vibration. The PWA shall evidence no degradation of function and meet the requirements of 3.3.1 thru 3.3.6 when subjected to the vibration test.

3.6.5 Burn-in. When required (see 6.2), the PWA shall be subjected to and pass a “power on” burn-in prior to the inspections listed in 4.1.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1).
b. Conformance inspections (CI) (see 4.1.2).

4.1.1 First article inspection. First article inspection shall be performed on the initial production-representative PWA of an order or when required by the contract. When a first article inspection is required (see 6.2 and 6.3), it includes all verifications listed in table II.

4.1.2 Conformance inspection. Conformance inspection shall include those examinations and tests from table II on all PWA produced as specified in the contract.

TABLE II. Verification methods.

Title	Requirements	Verification
Design, materials & manufacturing process	3.2	4.2
Operating requirements	3.3	4.2.3
Input power	3.3.1	4.2.3.1
CMOS logic levels	3.3.2	4.2.3.2
Data BUS isolation	3.3.3.1	4.2.3.3.1

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TABLE II. Verification methods - Continued.

Title	Requirements	Verification
I/O, MRD, & OEN-Q isolation	3.3.3.2	4.2.3.3.2
Display driver outputs	3.3.4	4.2.3.4
Clock clear	3.3.5.1	4.2.3.5.1
Q pulse	3.3.5.2	4.2.3.5.2
Clock count & latch	3.3.5.3	4.2.3.5.3
ATE test 1/	3.3.6	4.2.3.6
Interface requirements	3.4	4.2.4
PWA envelope	3.4.1	4.2.4.1
Ownership & support requirements	3.5	4.2.5
Interchangeability	3.5.1	4.2.5.1
Identification marking	3.5.2	4.2.5.2
Operating environment requirements	3.6	4.2.6
Temperature	3.6.1	4.2.6.1
Fungus	3.6.2	4.2.6.2
Shock	3.6.3	4.2.6.3
Vibration	3.6.4	4.2.6.4

1/ Not required unless specified (see 6.2).

4.2 Verification methods. The types of verification methods included in this section are visual inspection, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously-approved or previously-qualified designs.

4.2.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost-effective sampling procedures, to verify performance. See the contract for alternatives that replace verifications required by this specification.

4.2.2 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed under the following standard (room) ambient conditions:

a. Temperature: 77 ± 18°F (25 ±8°C).
b. Relative humidity: Uncontrolled room ambient.
c. Atmospheric pressure: Site pressure.

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4.2.3 Operating requirements verification.

4.2.3.1 Input power. To determine conformance to 3.3.1, the input current shall be measured with an ammeter.

4.2.3.2 CMOS logic levels. To determine conformance to 3.3.2, the input and output CMOS logic levels shall be observed and verified with an oscilloscope.

4.2.3.3 Input isolation.

4.2.3.3.1 Data bus isolation. To determine conformance to 3.3.3.1, the following procedures shall be performed:

a. Connect a 10 ± 10% kilohm (kohm) resistor to each of the bus board pins (P1-21 thru P1-28). Terminate each of the 10 kohm resistors with a 20 ± 10% kohm resistor to 5 V.
b. Apply 5 V to the connection between the 10 kohm and the 20 kohm resistors (the test input) connected to P1-21. Apply 0.00 V to the other seven test inputs. Verify that the voltage at P1-21 is not less than 4.5 V and the voltages at the other seven bus board pins are not greater than 0.5 V.
c. Repeat part b for the seven remaining board pins P1-22 thru P1-28.

4.2.3.3.2 I/O, MRD/, and OEN-Q isolation. To determine conformance to 3.3.3.2, the following procedures shall be performed:

a. Connect a 10 ± 10% kohm resistor to each of the board pins P1-29 thru P1-34, P1-36, and P1-38.
b. Apply 5.00 V to the unterminated end (the test input) of the 10 kohm resistor connected to P1-29. Apply 0.00 V to the other seven test inputs.
c. Verify that the voltage at P1-29 is not less than 4.5 V and the voltages at the other seven test inputs are not greater than 0.5 V max.
d. Repeat parts b and c for P1-30 thru P1-34, P1-36 and P1-38.

4.2.3.4 Display driver outputs. To determine conformance to 3.3.4, the following procedures shall be performed:

a. Connect a 270 ± 5% ohm resistor to each of the 32 display driver output board pins, P1-49 thru P1-80. Connect the unterminated end of each 270 ohm resistor to 5.00 V.
b. Write a hexadecimal bit pattern 00 to the bus (P1-21 thru P1-28) and apply the waveforms in figure 2 to latch the test data into U3, U4, U5, and U6. Verify that

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the voltages at all 32 display driver outputs (P1-49 thru P1-80) are not less than 2.75 V.

c. Write each of the bit patterns in table III to the bus and latch them into U3 by applying the waveforms in figure 3.

TABLE III. Display driver outputs.

BUS 7 (MSB)	BUS O (LSB)	Active output
0 0 0 0	0 0 0 1	P1-50
0 0 0 0	0 0 1 0	P1-53
0 0 0 0	0 1 0 0	P1-52
0 0 0 0	1 0 0 0	P1-56
0 0 0 1	0 0 0 0	P1-55
0 0 1 0	0 0 0 0	P1-51
0 1 0 0	0 0 0 0	P1-49
1 0 0 0	0 0 0 0	P1-54
0 0 0 0	0 0 0 0	------

d. Verify that the voltage at the active output listed in table III for each bit pattern is not greater than 1.00 V. Verify that each of the 31 display outputs not listed in the table for each bit pattern have a voltage of not less than 2.75 V.

e. Write each of the bit patterns in table IV to the bus and latch them into U4 by applying the waveforms in figure 4. Repeat part d.

f. Write each of the bit patterns in table V to the bus and latch them into U5 by applying the waveforms in figure 5. Repeat part d.

g. Write each of the bit patterns in table VI to the bus and latch them into U6 by applying the waveforms in figure 6. Repeat part d.

4.2.3.5 Real time clock.

4.2.3.5.1 Clock clear. To determine conformance to 3.3.5.1, 5.00 V shall be applied to both P1-38 and P1-33. U22 pin 10 shall be verified to maintain a logic level 1. U2 pins 3, 4, 5, 6, 11, 12, 13 and 14, and U21 pins 3, 4, 5, 6, 11, 12, 13, and 14 shall be verified to maintain a logic level 0.

4.2.3.5.2 Q pulse. To determine conformance to 3.3.5.2, a CMOS voltage level square wave (frequency between 50 kilohertz (kHz) and 410 kHz) shall be applied to P1-45, and 0.00 V to P1-43. The voltage level at P1-43 shall be brought to 5.00 V. The rising edge of the Q pulse at U16 pin 13 and P1-42 shall be verified to meet the timing restrictions of figure 1.

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4.2.3.5.3 Clock count and latch. To determine conformance to 3.3.5.3, apply the waveforms in figure 7 to clear and latch the real time clock count. The frequency of the signal applied to P1-45 shall be between 50 kHz and 410 kHz. The waveforms in figure 8 shall be applied to read the most and least significant bytes of the real time clock latch.

TABLE IV. Display driver outputs.

BUS 7 (MSB)	BUS 0 (LSB)	Active output
0 0 0 0	0 0 0 1	P1-58
0 0 0 0	0 0 1 0	P1-61
0 0 0 0	0 1 0 0	P1-60
0 0 0 0	1 0 0 0	P1-64
0 0 0 1	0 0 0 0	P1-63
0 0 1 0	0 0 0 0	P1-59
0 1 0 0	0 0 0 0	P1-57
1 0 0 0	0 0 0 0	P1-62
0 0 0 0	0 0 0 0	------

TABLE V. Display driver outputs.

BUS 7 (MSB)	BUS 0 (LSB)	Active output
0 0 0 0	0 0 0 1	P1-66
0 0 0 0	0 0 1 0	P1-69
0 0 0 0	0 1 0 0	P1-68
0 0 0 0	1 0 0 0	P1-72
0 0 0 1	0 0 0 0	P1-71
0 0 1 0	0 0 0 0	P1-67
0 1 0 0	0 0 0 0	P1-65
1 0 0 0	0 0 0 0	P1-70
0 0 0 0	0 0 0 0	------

TABLE VI. Display driver outputs.

BUS 7 (MSB)	BUS 0 (LSB)	Active output
0 0 0 0	0 0 0 1	P1-74
0 0 0 0	0 0 1 0	P1-77
0 0 0 0	0 1 0 0	P1-76
0 0 0 0	1 0 0 0	P1-80
0 0 0 1	0 0 0 0	P1-79
0 0 1 0	0 0 0 0	P1-75
0 1 0 0	0 0 0 0	P1-73
1 0 0 0	0 0 0 0	P1-78

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4.2.3.6 ATE test. To determine conformance to 3.3.6, install PWA into TACOM approved fixture, on TACOM approved, ATE equipment with TACOM approved software and instructions.

4.2.4 Interface requirements verification.

4.2.4.1 Overall envelope. Insert the PWA into the present existing location of the test set, verify its envelope and interfacing conformance.

4.2.5 Ownership and support requirements verification.

4.2.5.1 Interchangeability inspection. To determine conformance to 3.5.1, the interchangeability between different parts, subassemblies, and assemblies having the same military part number shall be verified.

4.2.5.2 Identification marking inspection. Verify the presence of the required PWA identification marking.

4.2.6 Operating environment requirements verification. The electrical performance tests specified in 4.2.3.1 through 4.2.3.5.3 shall be conducted prior to and at the completion of each environmental test. Verification of performance during the high temperature and low temperature environmental test shall be demonstrated by the tests specified in 4.2.3.3, 4.2.3.4, and 4.2.3.5.

4.2.6.1 Temperature. To determine conformance to 3.6.1, the PWA shall be tested as follows.

4.2.6.1.1 High temperature (operating). To determine conformance to 3.6.1.1, the PWA shall be subjected to high temperature test in accordance with method 501.3, procedure II (hot category) of MIL-STD-810, or equivalent (see 4.2.1), and tested to 4.2.3.1 thru 4.2.3.6.

4.2.6.1.2 Low temperature (operating). To determine conformance to 3.6.1.2, the PWA shall be subjected to low temperature test in accordance with method 502.3, procedure II of MIL-STD-810, or equivalent (see 4.2.1), and tested to 4.2.3.1 thru 4.2.3.6.

4.2.6.1.3 High temperature (nonoperating). To determine conformance to 3.6.1.3, the PWA shall be subjected to high temperature test in accordance with method 501.3, procedure I of MIL-STD-810, or equivalent (see 4.2.1).

4.2.6.1.4 Low temperature (nonoperating). To determine conformance to 3.6.1.4, the PWA shall be subjected to low temperature test in accordance with method 502.3, procedure I of MIL-STD-810, or equivalent (see 4.2.1).

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4.2.6.2 Fungus. To determine conformance to 3.6.2, the PWA shall be subjected to the fungus test in accordance with method 508.4 of MIL-STD-810 (28 day test), or equivalent (see 4.2.1).

4.2.6.3 Shock. To determine conformance to 3.6.3. The PWA shall be subjected to the shock test in accordance with method 516.4, procedure I, 40 g peak of MIL-STD-810, or equivalent (see 4.2.1), and tested to 3.3 after shock test.

4.2.6.4 Vibration. To determine conformance to 3.6.4, the PWA shall be subjected to a vibration test using a composite of figures 9 through 11. These figures represent the cargo environment at the cargo bed of a composite of tactical wheeled vehicles, the 5-ton M813 truck and the 12-ton M127 semitrailer. The data include differing vehicle loading conditions traversing over specially designed courses ranging from paved highway to off road conditions at various vehicle speeds. The spectrum is broadband random with peaks and notches at various discrete frequency bands. Break points are provided for establishing the spectrum shape. The test duration is 500 miles. The PWA shall be tested to 3.3 after the vibration test.

4.2.6.5 Burn-in. When required (see 6.2), the burn-in shall consist of three temperature cycles and a six gravity unit (g) root mean square (rms) random vibration as follows:

a. Temperature cycle. Each temperature cycle shall consist of 30 minutes at -60°F (-51°C) and 30 minutes at plus (+) 125°F (52°C). The rate of temperature change between temperature extremes shall be not less than 5°F (2.8°C) per minute. Power shall be applied to the PWA for not less than 30 minutes from 20°F (7°C) through the completion of each high-temperature soak.
b. Random vibration. Following all temperature cycling, the PWA shall be hard mounted to the vibration table so that the axis of vibration is perpendicular to the plane of the PWA. The vibration shall be at room ambient temperature, for a period of 10 minutes, as shown in figure 12.

NOTE: PWAs being procured for simultaneous installation in VTMs being procured do not have to be subjected to “burn-in” provided the VTMs are subjected to the 16-hour “power on” burn-in.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activity within the Military Department or Defense Agency, or within the Military

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Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The PWA contains a 16-bit resetable real time clock with latch and timing circuitry. High and low bytes of the latched clock data can be placed on BUS 0 - BUS 7 (board pins 21-28) independently. (Board pins will hereafter be referred to as P1-XX where XX = the pin number.) The clock count is incremented at one fourth the frequency of the signal at P1-45. The clock section also provides a Q pulse output (P1-42), a self-test output (P1-40), and an EF4 output (P1-39). The Display Driver consists of four latches (U3, U4, U5, U6) which can each latch 8 bits of data off BUS 0 - BUS 7. Each output of each latch drives the base input of an open collector darlington transistor pair. When a logic level 1 is present at the base input, the collector input of the darlington pair will sink current, lighting up a segment of an external display which is pulled up to +5 Vdc. Connected to each of the collector inputs is a 510 ohm resistor to ground which keeps the external display segment partially on when it is not being driven by the darlington pair. The PWA also provides an output from U22, pin 11 to P1-41. The PWA is military unique because it is designed specifically to be a component of the Simplified Test Equipment/Internal Combustion Engine - Reprogrammable (STE/ICE-R) Vehicle Test Meter (VTM), and to withstand environmental conditions which exceed commercial requirements.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. PWA drawing number, revision letter, and date and PWA part number (see 3.2 and 3.5.2).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. If first article inspection is required (see 3.1).
e. If CMOS logic levels should be other than as specified (see 3.3.2).
f. If marking is other than as specified (see 3.5.2).
g. If burn-in is not required (see 3.6.5).
h. If arrangements for first article inspection are other than as specified (see 4.1).
i. Whether ATE test is required (see table II footnote).
j. Packaging requirements (see 5.1).

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6.3 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.4 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, Government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from table II and their designated frequency based on a risk assessment for the procurement.

6.5 Historical data. Historically, ATPD 2148, dated 9 March 1988, was superseded by MIL-P-62739(AT).

6.6 Subject term (key word) listing.

Real time clock
Simplified Test Equipment/Internal Combustion Engine - Reprogrammable (STE/ICE-R)
Vehicle test meter (VTM)

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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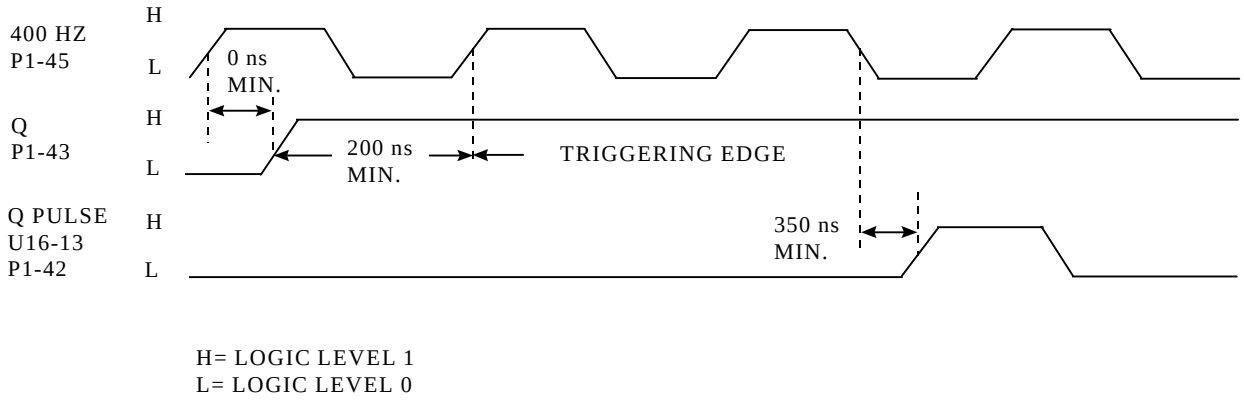


FIGURE 1. Q pulse timing.

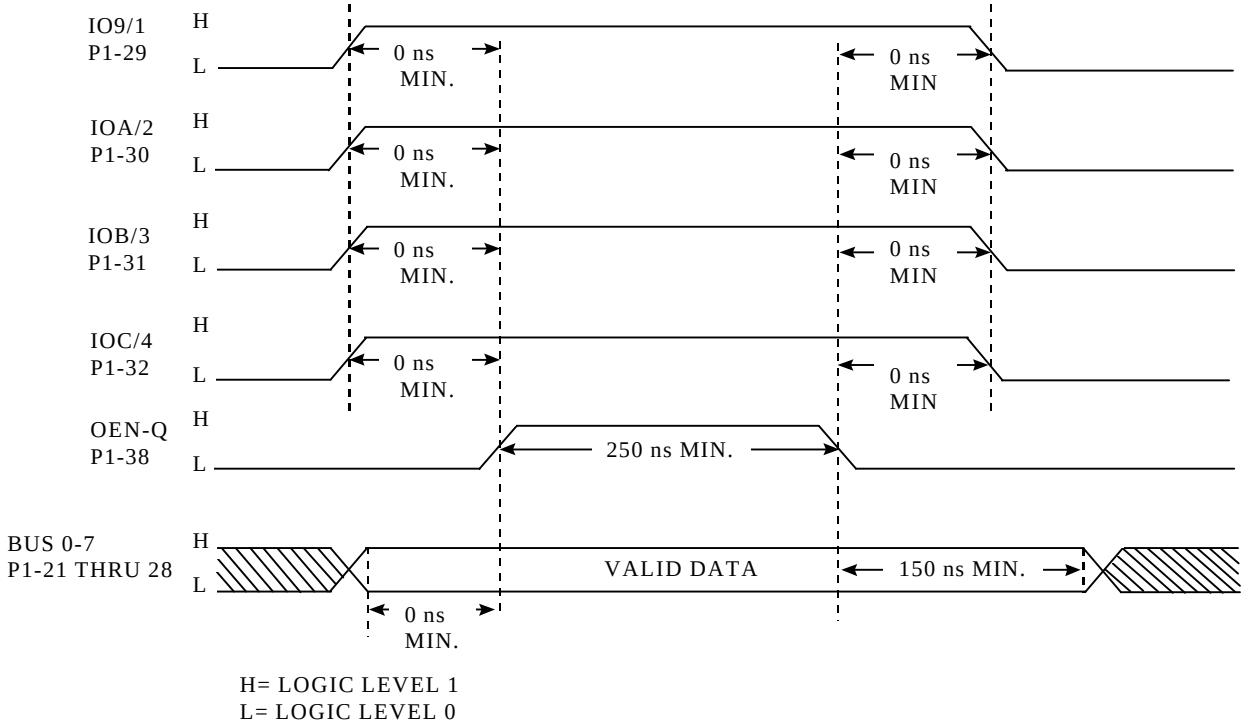


FIGURE 2. Waveforms for writing data to all display outputs.

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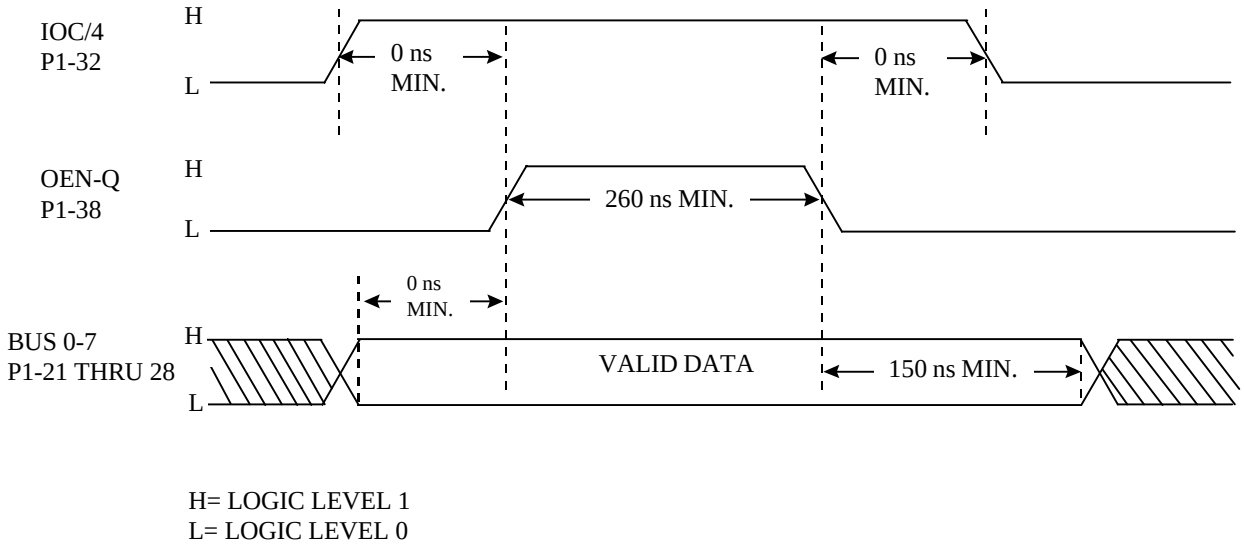


FIGURE 3. Waveforms for writing data to U3.

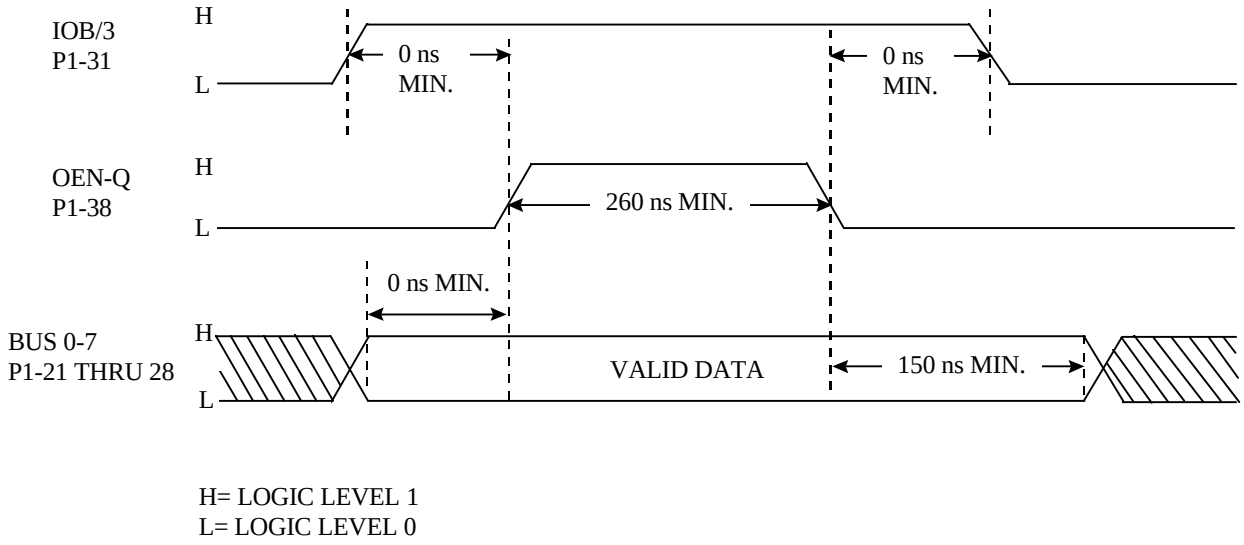


FIGURE 4. Waveforms for writing data to U4.

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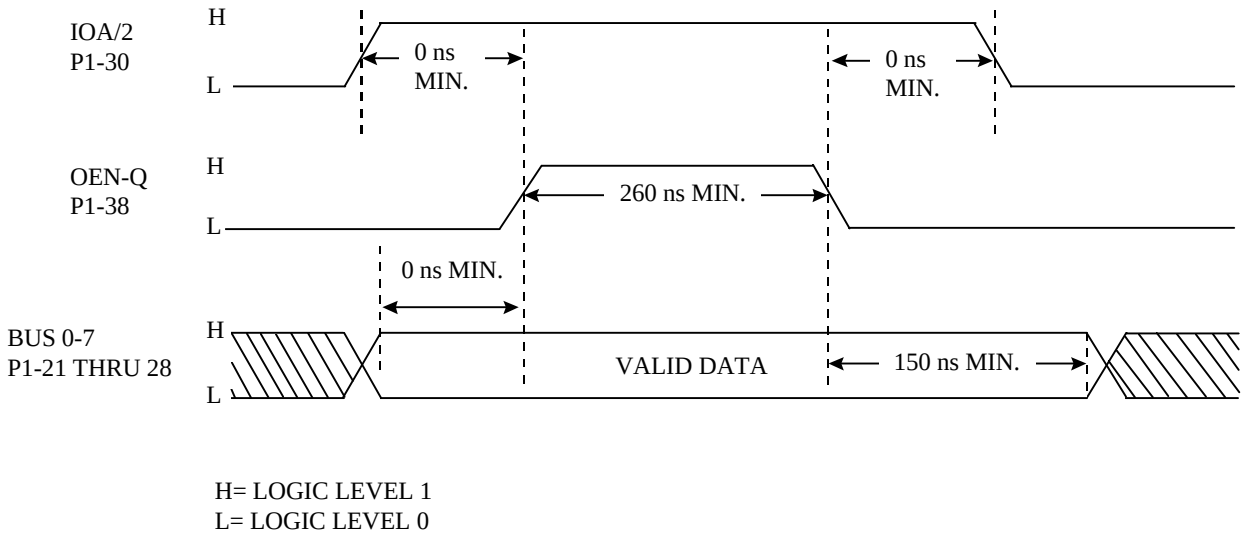


FIGURE 5. Waveforms for writing data to U5.

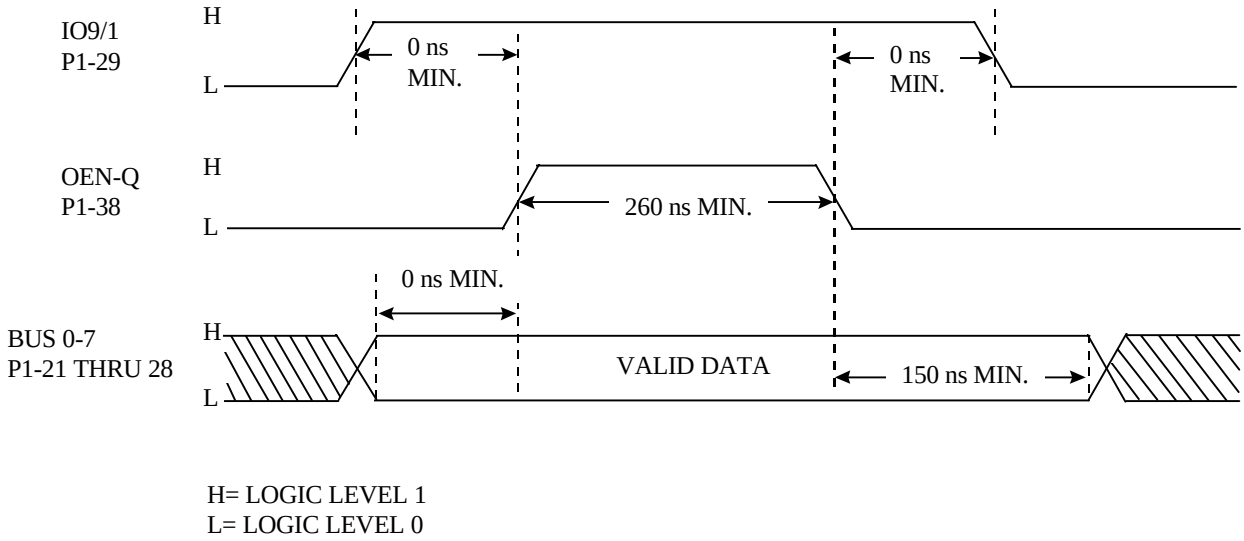


FIGURE 6. Waveforms for writing data to U6.

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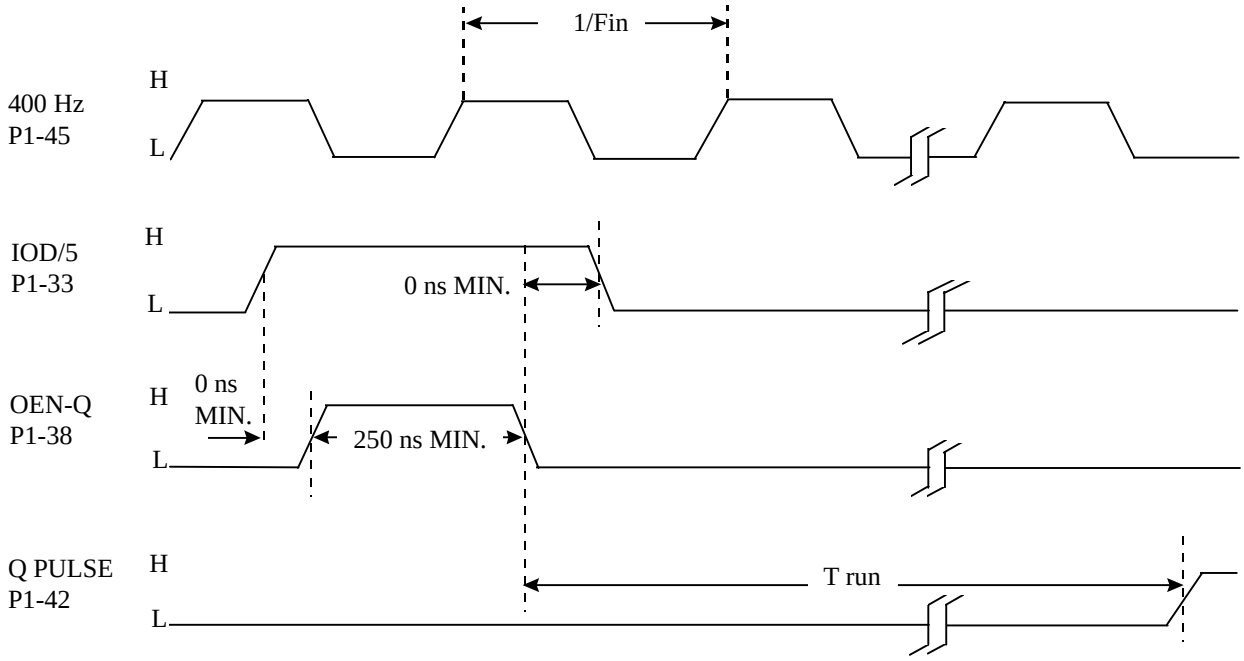


FIGURE 7. Waveforms to clear and latch real time clock.

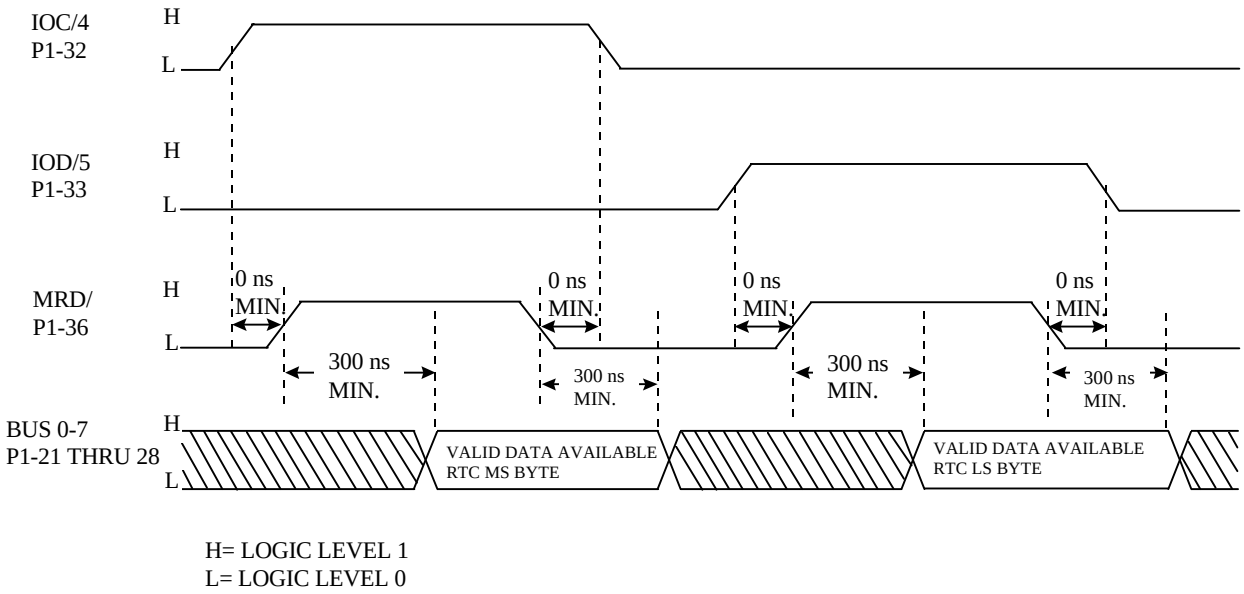
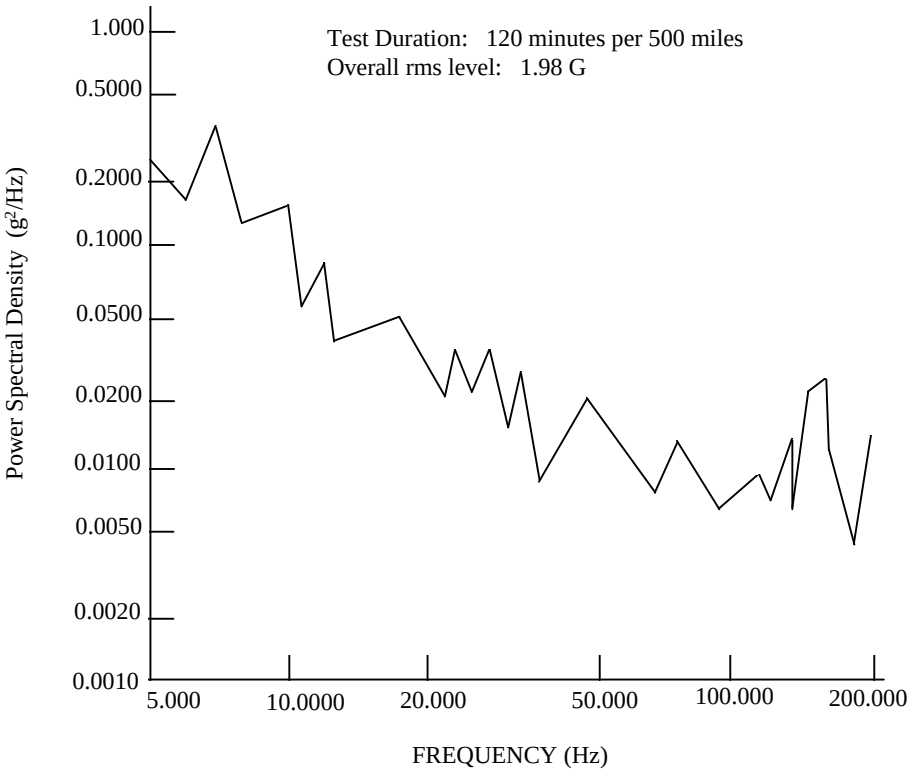


FIGURE 8. Waveforms to read real time clock latches.

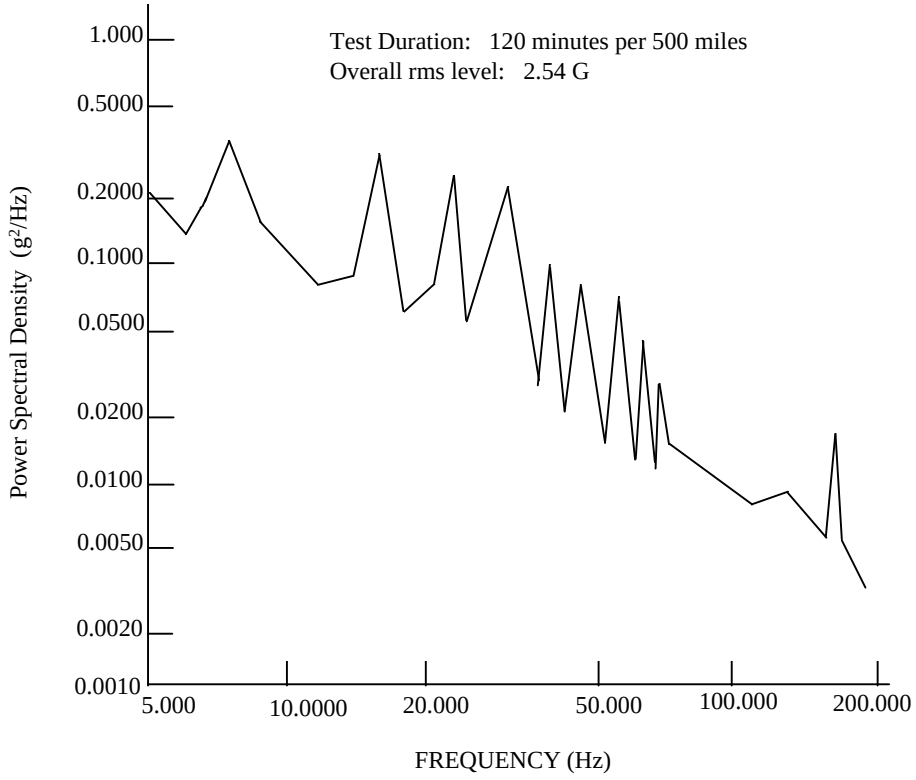
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BREAKPOINTS	
FREQ	PSD VALUE
5	0.23840
6	0.16840
7	0.38230
8	0.14870
10	0.16840
11	0.05930
12	0.09750
13	0.04090
18	0.05370
23	0.01940
24	0.03520
26	0.02310
28	0.03520
32	0.01670
33	0.03110
37	0.00810
49	0.02190
63	0.00700
73	0.01150
98	0.00660
120	0.00970
123	0.00680
138	0.01340
140	0.00760
146	0.02490
159	0.02820
163	0.01300
188	0.00480
200	0.01340

FIGURE 9. Basic transportation, composite tactical wheeled environment, vertical axis.

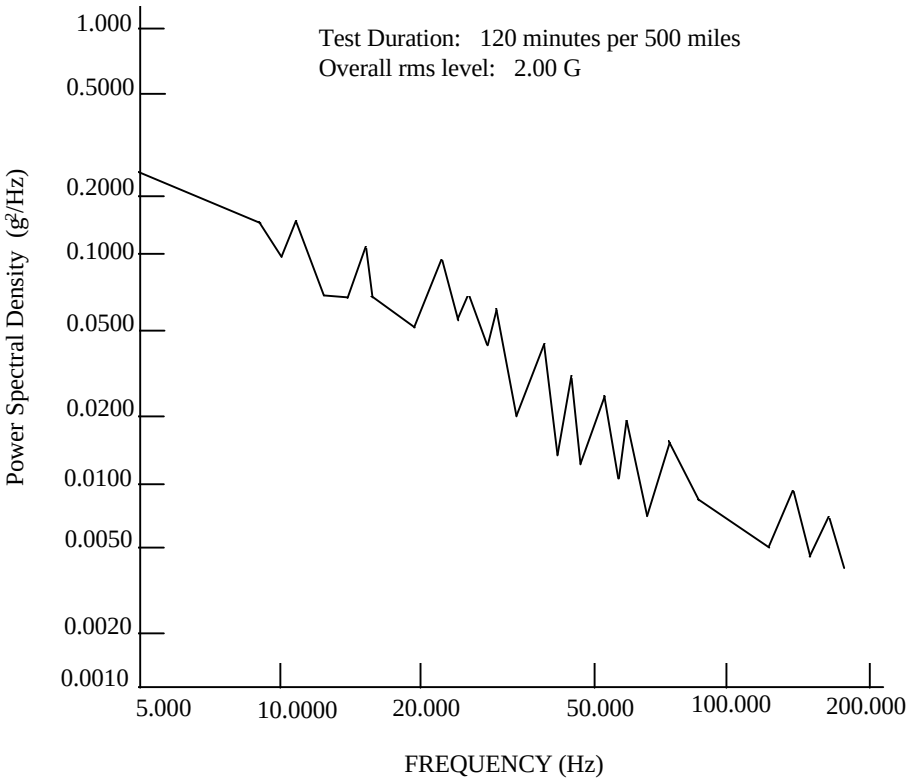
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BREAKPOINTS	
FREQ	PSD VALUE
5	0.22130
6	0.13140
7	0.20540
8	0.38230
9	0.15630
12	0.07990
15	0.08830
16	0.32940
19	0.06080
22	0.09280
24	0.26340
27	0.05640
32	0.22690
36	0.02820
40	0.11600
44	0.02090
48	0.08190
53	0.01480
56	0.07420
60	0.01270
64	0.04980
65	0.01100
68	0.03260
69	0.01480
114	0.00740
132	0.00830
160	0.00510
169	0.01710
173	0.00550
200	0.00310

FIGURE 10. Basic transportation, composite tactical wheeled environment, longitudinal axis.

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BREAKPOINTS	
FREQ	PSD VALUE
5	0.25490
10	0.13030
11	0.09670
12	0.13360
13	0.06500
15	0.06340
16	0.11230
17	0.06500
22	0.04590
24	0.08980
26	0.05070
28	0.06830
30	0.03670
32	0.05880
38	0.01920
40	0.03860
44	0.01290
48	0.02940
50	0.01140
56	0.02460
60	0.00960
63	0.01880
70	0.00660
78	0.01500
93	0.00730
137	0.00460
145	0.00910
162	0.00430
176	0.00660
200	0.00360

FIGURE 11. Basic transportation, composite tactical wheeled environment, transverse axis.

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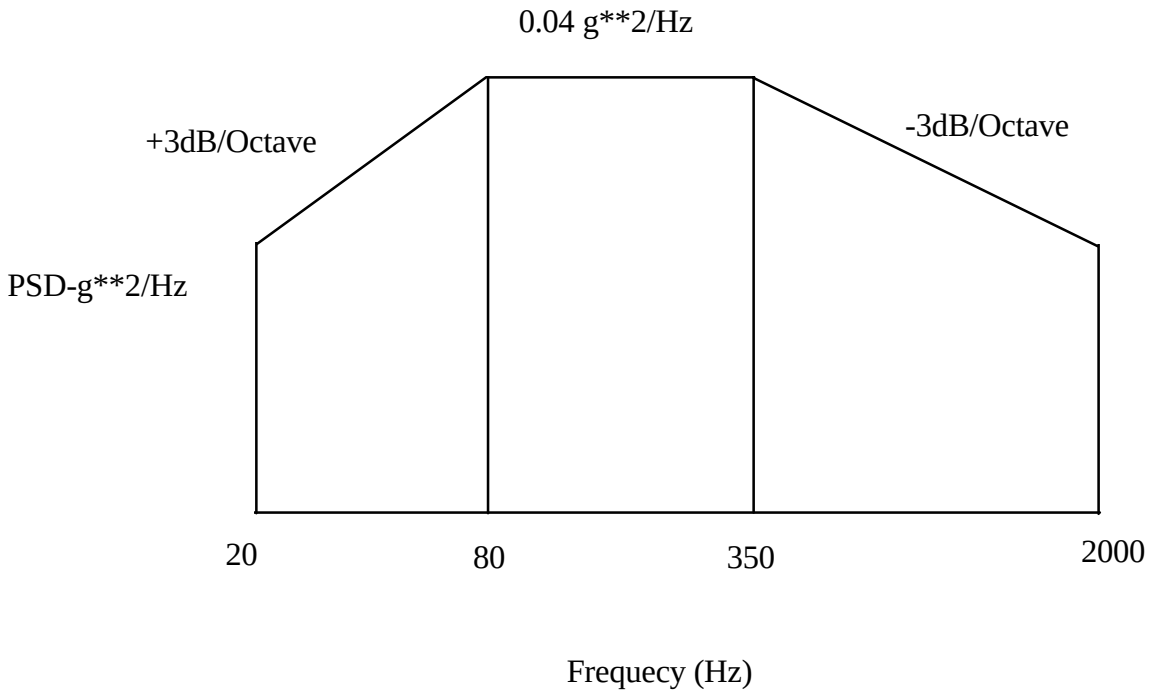


FIGURE 12. Random vibration spectrum.

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Army - AT

Preparing Activity:
Army - AT

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c. ADDRESS *(Include Zip Code)* Commander U.S. Army Tank-automotive and Armaments Command ATTN: AMSTA-TR-E/BLUE Warren, MI 48397-5000	(810) 574-8745 786-8745 IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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